Mechanics Final Project: Logistic Map

[redacted]

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1 Preliminary Background Section

Logistic maps are useful when studying the growth of a population by modelling how it could grow, or when modeling a predator prey relationship by seeing what different actions could be taken. For equations of this type, there is typically an initial condition such as x_0 and a changeable variable such as r, with r being greater than 0. With this type of equation, you can then plot out the different ways this system can evolve depending on that r component, producing a sequence of x value such that it outputs x_0, x_1, x_2, x_3[1].

The basic example of a logistic map would be the equation

$$n_{t+1} = rn_t(1 - \frac{n_t}{N}) \tag{1}$$

where n_{t+1} is each successive value of n starting at n_1, and n_t is the previous value. In this case n_t would be an initial value n_0, and N is a positive number that signifies the maximum value that n can reach. Or in other terms, N is the carrying capacity of the system. An interesting note about the logistic map is that it is nonlinear, which allows for chaotic behavior to occur, whereas an exponential map does not[1].

For the purposes of the upcoming problem, simplifying the equation will help with understanding how it works. This equation can be simplified using the relation $x = \frac{n}{N}$, so that the equation simplifies to

$$x_{t+1} = rx_t(1 - x_t)[1]. \tag{2}$$

To find the fixed points of the system, we set the equation equal to x,

$$x = rx(1 - x) \tag{3}$$

which can then be manipulated into

$$0 = rx - rx^2 - x \tag{4}$$

thus, the fixed points of this example are $x = 0$ and $x = \frac{r-1}{r}$.

To determine how stable these fixed points are, we first take the derivative of the equation, which in this case would be

$$r - 2rx[1]. \tag{5}$$

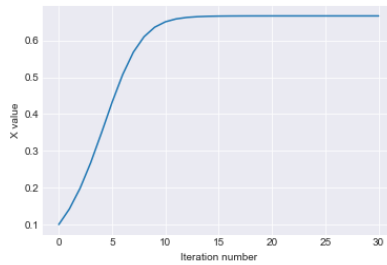

Figure 1: This is a graph of the equation for x_0=0.1 and r=1.5, after 10 iterations the growth begins to level off.

After which, the stability points are substituted into the derived equation one at a time to determine at what value r would need to be for the point to be stable, that is, when does $f'(point) > 1$. These are the core concepts that will be needed in order to develop an understanding of how logistic maps can be used, as will be displayed in the problem below.

2 The Problem

2.1 Part A

For this project, the logistic equation $x_{n+1} = rx(1 - \frac{x}{2}$ was given as the starting point. First, we needed to plot the value of x over it's first 30 iterations using the initial x value of 0.1, and an r-value of 1.5. This graph was made using python and can be seen in Figure 1. To check to see whether this graph makes sense, we can algebraically solve for the fixed points of the equation and see what the stability of those points are. To begin, we can use the method discussed in the background and set the equation equal to x

$$x = rx(1 - \frac{x}{2}). \tag{6}$$

Solving for the stability points gives an x-value of 0 and 2-$\frac{2}{r}$, which is equivalent to 0.67. To check for the stability of these points, we first find the derivative of the equation as discussed in the background, in this case it is $2r - 2rx$. Setting this equation equal to 1, and knowing that r=1.5, and the fixed x values are either 0 or 0.67, the stability of the points can be evaluated as follows. For x=0,

$$2r - 2rx => 2r > 1 \tag{7}$$

so when $r < 0.5$, the fixed point x=0 is stable. For x=0.67,

$$2r - 1.34r = 1 => 0.66r = 1 => r = 1.52 \tag{8}$$

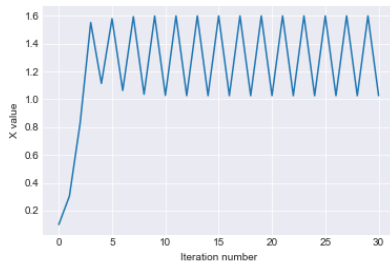

Figure 2: This is a graph of the equation for x_0=0.1 and r=3.2, after a few iterations the system begins to oscillate between two fixed points.

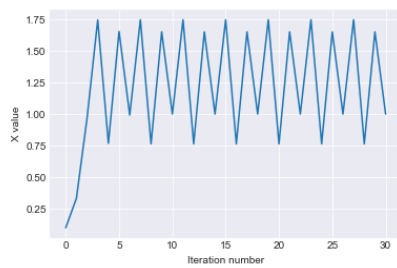

Figure 3: This is a graph of the equation for x_0=0.1 and r=3.5, after 10 iterations the growth begins to oscillate between 4 different fixed points.

so when $r < 1.52$, the fixed point is stable, which can be seen in Figure 1. At x=0, the growth of the equation continues to increase, but it begins to level off at x=0.67. This fits with our analytical solution as the r-value for this system is less than 1.51.

2.2 Part B

For the next part of this problem, the r-value was first increased to 3.2, with the results being displayed in Figure 2. The r-value is then increased to 3.5, with the subsequent graph being the graph in Figure 3. This allows for a look at what happens as the r value increases for this equation. In Figure 2, the system is only oscillating between two different points after 5 iterations of the equation. In Figure 3 however, the system is now oscillating between 4 different fixed points after around 4 iterations. This indicating that as the r value increased, the number of fixed points in the system will also increase, which can lead to chaotic situations.

```python
def logisticnew(r,x):
    return r*x*(1-(x/2))

n = 10000
r = np.linspace(2.8, 4.0, n)

iterations = 1000
last = 100


fig, ( ax1) = plt.subplots( 1, figsize=(8, 9),
                            sharex=True)
for i in range(iterations):
    x = logisticnew(r, x)
    # We display the bifurcation diagram.
    if i >= (iterations - last):
        ax1.plot(r, x, ',k', alpha=.25)
ax1.set_xlim(2.8, 4)
ax1.set_title("Bifurcation diagram")
plt.xlabel('r values from 2.8 to 4.0')
plt.ylabel('x values')
```

Figure 4: The code used to create the bifurcation diagram in Figure 5.

2.3 Part C

After the fixed points have been determined for the system when r=1.5,3.2, and 3.5, the next step is to investigate what a bifurcation diagram for an x_0 value of 0.1 and a range of r values from 2.8 to 4.0 would look like, which was done using the code shown in Figure 4[3].

From the bifurcation diagram, it can be seen that the system has a long term constant behavior, until around r-value 3.55, and then begins to behave chaotically until a little after r-value 3.7 and then again a little before r-value 3.8. This happens again a little after r-value 3.8, with there being multiple periods that emerge and then branch back off into chaos.

2.4 Part D

From Figure 5, it can be seen that the rate at which the period starts doubling is increasing as time goes on. This can be investigated using the Feigenbaum relation

$$(\gamma_{n+1} - \gamma_n) \approx \frac{1}{\delta}(\gamma_n - \gamma_{n-1}). \tag{9}$$

Looking at this equation, if we replace the γ with a r, we can then state that

$$(r_{n+1} - r_n) \approx \frac{K}{\delta^n} \tag{10}$$

where K represents $r_n - r_{n-1}$, and the delta represents the rate at which doubling is occurring for the system.

Using the bifurcation diagram as a guide, the r values when the first, second, and third doubling occur are 3.000, 3.450, and 3.570. Using these values and

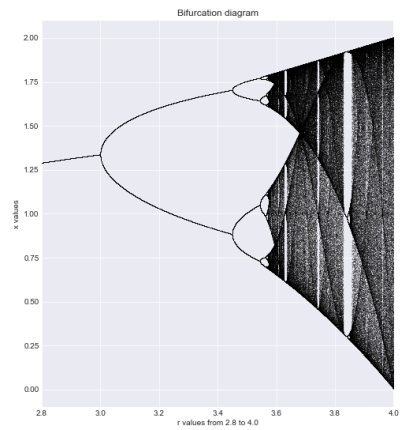

Figure 5: This is the bifurcation diagram for r values of 2.8 to 4.0 and a x_0 value of 0.1.

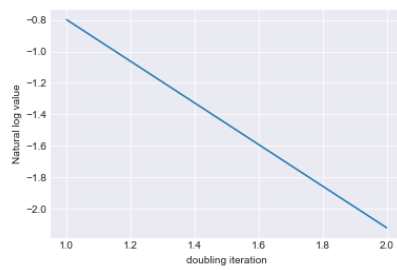

Figure 6: The log value compared to what iteration of doubling has occured.

their equivalent n values, a graph like that in Figure 6 can be created. This is done by taking the natural log of the first part of the above equation, such as

$$ln(3.450 - 3.000) \tag{11}$$

and the same for the third value minus the second value. After which these two values are plotted with respect to their respect n values of 1 and 2. The slope of this graph is the delta value for the system, which represents how fast the system will begin to double[3]. For this system, the delta is 1.37, so this is how fast the period doubling cascade will occur as r varies.

3 Citations

1. Taylor, John R. Classical Mechanics. University Science Books, 2005.
2. Rembold, Jed. "The Logistic Map."
3. "12.1. Plotting the Bifurcation Diagram of a Chaotic Dynamical System." IPython Cookbook, Second Edition, ipython-books.github.io/121-plotting-the-bifurcation-diagram-of-a-chaotic-dynamical-system/.